

Data Mining

Hierarchical Clustering-BIRCH

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Advantages of BIRCH vs. Other Clustering Algorithms

- ▶ BIRCH is "local"
 - clusters a point without having to check against all other data points or clusters
- ▶ Can remove outliers ("noise")
- ▶ Produce good clusters with a single scan of dataset
- ▶ Linearly scalable
 - minimizes running time
 - adjusts quality of result with regard to available memory

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BIRCH: Key Components

▶ Clustering Feature (CF)

- Summary of the statistics for a given cluster: the 0-th, 1st and 2nd moments of the cluster from the statistical point of view
- Used to: compute centroids, and measure the compactness and distance of clusters

▶ CF-Tree

- height-balanced tree
- two parameters:
 - number of entries in each node
 - The diameter of all entries in a leaf node
- Leaf nodes are connected via *prev* and *next* pointers

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Clustering Feature

Clustering Feature (CF): $CF = (N, LS, SS)$

N : Number of data points

LS : linear sum of N points: $\sum_{i=1}^N X_i$

SS : square sum of N points: $\sum_{i=1}^N X_i^2$

$CF_3 = CF_1 + CF_2 = (3+3, (9+35, 10+36), (29+417, 38+440)) = (6, (44, 46), (446, 478))$



$CF_1 = (3, (2+3+4, 5+2+3), (2^2+3^2+4^2, 5^2+2^2+3^2)) = (3, (9, 10), (29, 38))$

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Properties of Clustering Feature

- ▶ CF entry is a **summary** of statistics of the cluster
- ▶ A **representation** of the cluster
- ▶ A CF entry has **sufficient information** to calculate the centroid, radius, diameter and many other distance measures
- ▶ **Additively** theorem allows us to **merge** sub-clusters **incrementally**

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Distance Measures

□ Given a cluster with data points

□ Centroid:

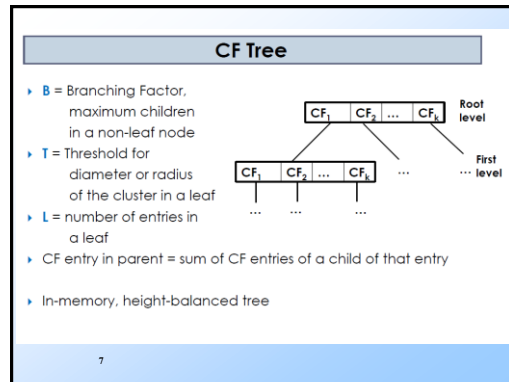
$$x_0 = \frac{\sum_{i=1}^n x_i}{n}$$

□ Radius = R = Maximum distance between all the points and the centroid

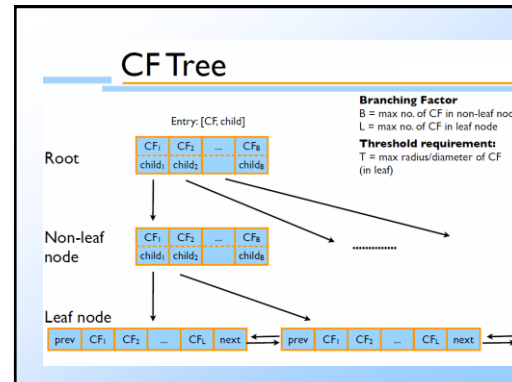
□ Diameter = D = Maximum distance between any two points of the cluster

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Properties of Different Clustering Techniques - Summary

- Description of different clustering algorithms in terms of (i) shapes of the clusters that can be determined; (ii) input parameters that must be specified and (iii) limitations.
- **K-Mean**
 - (i) Spherical-shaped clusters; (ii) The number of clusters; (iii) Sensitive to noise and outliers. Works well on small data sets only but can be scaled.
- **K-Medoids**
 - (i) Spherical-shaped clusters; (ii) The number of clusters; (iii) Work only on small data sets (not scalable).

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- **CLARA**
 - (i) Spherical-shaped clusters; (ii) The number of clusters; (iii) Sensitive to the selection of initial samples.
- **CLARANS**
 - (i) Spherical-shaped clusters; (ii) The number of clusters; (iii) Sensitive to the sampling techniques used.
- **BIRCH**
 - (i) Spherical-shaped clusters (uses radius and diameter); (ii) N d -dimensional data points; (iii) Because a CF tree can hold only a limited number of entries due to its size, a CF tree does not always correspond to what a user may consider a natural cluster.

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Evaluation of Clustering

Major tasks of clustering evaluation are

- **Assessing clustering tendency**
 - Assess whether a nonrandom structure exists in the data.
 - Clustering is meaningful only when there is a nonrandom structure in the data.
 - Can use different statistical method to check randomness such as **Hopkins Statistic**.

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Evaluation of Clustering

- **Determining the number of clusters in a data set.**
 - Often depends on the distribution's shape and scale in the data set, as well as the clustering resolution required by the user
 - Some simple methods are
 - (i) Set the number of clusters to about $\sqrt{\frac{n}{2}}$, so that the expected number of points in each cluster is $\sqrt{2n}$
 - (ii) Use the Elbow Method

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Evaluation of Clustering

□ *Measuring clustering quality*

- Assess how good the resulting clusters are

We have different methods based on whether ground truth are available or not.

Extrinsic Methods (when ground truth are available, also known as supervised methods)

Intrinsic Methods (when ground truth are not available, also known as unsupervised methods)

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